Julian Karlbauer M.Sc.

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My research interests lie at the intersection of spatial computing, digital fabrication, and intelligent interactive systems, with a focus on empowering and augmenting unique human abilities using an applied, systems-focused, and interdisciplinary approach.

EXPERIENCE

Sabbatical *Latin America, Middle East, Europe, Asia (06/2025 – present)*

Planned sabbatical focused on gaining perspective through cross-cultural engagement, journalism, independent projects, introspection, and clarifying my long-term trajectory as a researcher.

Phygtl Inc. *Founding Engineer* *Palo Alto, USA (remote) (12/2024 – 06/2025)*

Phygtl is a Silicon Valley based startup dedicated to bridging physical and digital spaces through collaborative social AR experiences and enabling Generative-AI based 3D co-creation through real-world visual input. My role as a Founding Engineer was to architect and build an AR frontend for iOS & Android devices, with a strong focus on UI/UX and efficient software architecture. I collaborated directly with the executive team (CEO, CTO) to define the technical roadmap.

Ulm University *Research Associate* *Ulm, Germany (01/2024 – 10/2024)*

Worked as a Research Associate under Prof. Dr. Enrico Rukzio on *Personal Fabrication*. My focus was on Generative-AI-supported 3D authoring in AR to enhance Digital Manufacturing workflows. As part of my work, I attended ACM SCF '24 and demoed *MixGen*, a working AR prototype for generative 3D content authoring from multi-modal real-world input. I further attended ACM SCF Summer School and co-published *VRCReatIn*, a tablet-based 3D content authoring VR tool, at ACM MUM '24 [3].

Siemens AG *Master's Thesis Student* *Munich, Germany (04/2023 – 11/2023)*

I conducted my Master's Thesis on enhancing Quality Assurance in Additive Manufacturing processes through data-driven Digital Twin and Augmented Reality Interfaces. I designed, built, and evaluated an AR interface to an industrial metal 3D printer (WAAM), enabling interactive and in-situ visualisation of Digital Twin data, such as CAD and spatio-temporal anomaly prediction. The prototype received high institutional validation at Siemens and was recognised as a future candidate for demonstration to the CEO and at the Formnext exhibition.

Audi AG *Software Engineering Intern* *Ingolstadt, Germany (03/2022 – 09/2022)*

Worked in the HCI team of the Audi Production Lab to develop a mobile cross-device Augmented Reality platform designed for Audi internal communication on 3D Digital Twin data. I was responsible for iOS-facing Unity3D development, Interaction Design, and assistance in conducting interviews and user-studies.

TriCAT GmbH *R&D Intern* *Ulm, Germany (03/2021 – 06/2021)*

Proposed and built a machine vision research prototype for in-situ real-time object recognition and 3D localisation on HoloLens 2 using manually fine-tuned CNNs and environment meshes with a compute-efficient remote GPU architecture. Through my work, I pioneered applying Deep Learning tools for problem-solving in the company.

Ulm University *Research Assistant* *Ulm, Germany (04/2018 – 03/2021)*

Significantly contributed to rigorous research published in ACM CHI and ACM ToCHI, with a focus on 3D authoring [1] and health aspects in Virtual Reality [2]. My responsibilities included XR prototyping, contributing to design processes, providing novel ideas to research projects, conducting user studies, creating footage and media for publications.

EDUCATION

Ulm University *M. Sc. in Media Informatics* *Ulm, Germany (10/2020 – 03/2024)*

Thesis Title: *"Show, Don't Tell: Enhancing Quality Assurance in Additive Manufacturing through Augmented Reality"*
(Thesis Grade 1.0; Overall Grade: 1.2)

Norwegian University of Science and Technology *Visiting Student* *Trondheim, Norway (08/2021 – 12/2021)*

Thesis Title: "*Raw Audio Music Emotion Composition using Mood Annotated Spotify Playlists*"
(Thesis Grade 1.0; Overall Grade: 1.0)

Ulm University *B. Sc. in Media Informatics* *Ulm, Germany (04/2016 – 07/2020)*

Thesis Title: *"An Exploration of Foveated Blue Light Filters in Virtual Reality"*
(Thesis Grade 1.0)

ACADEMIC SERVICES

- Conference Reviewer CHI 2025
- Served as a student volunteer at the ACM SCF 2024 conference.

ACADEMIC PUBLICATIONS

[1] Tobias Drey, Nico Rixen, **Julian Karlbauer**, and Enrico Rukzio. 2024. VRCreatIn: Taking In-Situ Pen and Tablet Interaction Beyond Ideation to 3D Modeling Lighting and Texturing. In Proceedings of the International **Conference on Mobile and Ubiquitous Multimedia (MUM '24)**. Association for Computing Machinery, New York, NY, USA, 24–35. https://doi.org/10.1145/3701571.3701580

[2] Teresa Hirzle, Fabian Fischbach, **Julian Karlbauer**, Pascal Jansen, Jan Gugenheimer, Enrico Rukzio, and Andreas Bulling. 2022. Understanding, Addressing, and Analysing Digital Eye Strain in Virtual Reality Head-Mounted Displays. **ACM Trans. Comput.-Hum. Interact.** 29, 4, Article 33 (August 2022), 80 pages. https://doi.org/10.1145/3492802

[3] Tobias Drey, Jan Gugenheimer, **Julian Karlbauer**, Maximilian Milo, and Enrico Rukzio. 2020. VRSketchIn: Exploring the Design Space of Pen and Tablet Interaction for 3D Sketching in Virtual Reality. In Proceedings of the 2020 **CHI Conference on Human Factors in Computing Systems (CHI '20)**. Association for Computing Machinery, New York, NY, USA, 1–14. https://doi.org/10.1145/3313831.3376628